

Question	Marking Guidance	Mark	Comments
04.1	1. <u>Lower</u> (force of contraction) in mouse/B (than control/100%) below 29 °C OR <u>Lower</u> (force of contraction) in rabbit/D (than control/100%) below 26.5 °C; 2. <u>Higher</u> (force of contraction) in mouse/B (than control/100%) above 29 °C OR <u>Higher</u> (force of contraction) in rabbit/D (than control/100%) above 26.5 °C; 3. Only (used) mouse and rabbit OR No other organism/species used; 4. Body temperature of mouse/rabbit higher (than temperatures investigated); 5. Only used one/0.5 pH (below typical pH) OR (Should) use more pH values; 6. (Used) isolated muscle tissue; 7. No stats test to see if (difference is) <u>significant</u>;	4 max	1. Accept any temperature below 29 °C for mouse/B or any specified temperature below 26.5 °C for rabbit/D. 2. Accept any temperature above 29 °C for mouse/B or any temperature above 26.5 °C for rabbit/D. 1. and 2. Accept 27 °C for 26.5 °C and accept 28.5 °C for 29 °C. 3. Accept only two animals/species used. 4. Accept body temperature of mouse/rabbit not known 7. Ignore SD.
04.2	1. (Less/No) <u>tropomyosin</u> moved from binding site OR Shape of <u>tropomyosin</u> not changed so binding site not exposed/available; 2. (Fewer/No) actinomyosin bridges formed; 3. Myosin head does not move OR Myosin does not pull actin (filaments) OR (Less/No) <u>ATP</u> (hydrol)<u>ase</u> (activation);	3	1 and 2. Reject active site only once. 1. Ignore troponin. 2. Accept actin and myosin do not bind. 3. Reject ATP synthase. Do not penalise reference to calcium rather than calcium ions. Credit all mark points even if context relates to what happens when calcium ions are present.

04.3	1. Regenerates/produces NAD OR oxidises reduced NAD; 2. (So) glycolysis continues;	2	1. Reject NADP and any reference to FAD. 1. Accept descriptions of oxidation e.g. loss of hydrogen. 2. Accept description of glycolysis e.g. glucose to pyruvate. 2. Accept 'for oxidising/converting triose phosphate to pyruvate'.
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